



Department of Education

STEM Education
National Schools of Excellence
STEM Chemistry Syllabus
Grade 12
Term 2

DOMAIN **Phylum**

SCIENCE **Chemistry**

Technology

Engineering

Mathematics

Issued free to schools by the Department of Education

Published in 2021 by the Department of Education, Papua New Guinea

© Copyright 2021, Department of Education, Papua New Guinea

All rights reserved. No part of this publication may be reproduced, stored in a retrieval system or transmitted by any form or by any means electronic, mechanical, photocopying, recording or otherwise without the prior written permission of the publisher.

ISBN

Acknowledgements

Ultima Thulë Ltd was engaged by the Department of Education to develop the STEM Curriculum. The content was generated by writers from universities, secondary schools, new university graduates with knowledge in the subject matter and industry subject experts. Quality assurance groups and the Science Subject Advisory Committee have also contributed to the development of this syllabus.

This document was developed with support from the General Education Services Division of the Department of Education.

Table of Contents

SECRETARY'S MESSAGE.....	4
INTRODUCTION.....	5
RATIONALE.....	6
STEM DOMAIN	7
STEM CHEMISTRY CONTENT OVERVIEW – GRADE 12, TERM 2.....	7
Table 1.1 Sequencing & Benchmarking STEM Chemistry Content Standards.....	8
Table 1.2 Mapping Content Progressions by Modules.....	10
THEMA: ENERGY	11
MODULE 12.1: HYDROGEN – FUTURE OF ENERGY	13
Topic 1: Hydrogen as Fuel.....	14
Topic 2: Methods of Producing Hydrogen	15
Topic 3: Storing Hydrogen	17
Topic 4: Hydrogen Fuel cell.....	18
THEMA: BATTERY TECHNOLOGY	19
MODULE 12.1: SOLID-STATE BATTERY.....	21
Topic 1: Introduction to Solid-State Battery.....	22
Topic 2: Solid Electrolyte.....	23
Topic 3: Manufacturing of Solid Batteris	24
REFERENCES	25

SECRETARY'S MESSAGE

This Chemistry Syllabus is to be used exclusively by teachers of Chemistry to teach Grade 11 Students enrolling for the STEM Education through the Schools of Excellence throughout Papua New Guinea. The STEM Chemistry Bridging Syllabus ensured that Grade 11 students enrolling for STEM Education were grounded in the basic concepts espoused in Chemistry before undertaking the advanced concepts introduced in this syllabus.

The syllabus conforms to the visions of the National Education Plan and the School of Excellence Policy. Furthermore, it is undergirded by a policy and legislative matrix that were issued by the National Government over the last decades, such as the Vision 2050 and Development Strategic Plan 2030. The Preamble of the National Constitution, especially the First Directive of the *National Goals and Directive Principles* makes a clarion call for the Integral Human Development, which the Department took heed of to ensure, through this new syllabus, that *all forms of beneficial creativity, including sciences and cultures, [are] to be actively encouraged*. A principal objective of the *National Education Act 1983 (amended 1995)* reiterates this virtue of Integral Human Development (Section 4).

In accordance with the widespread demand for a home-grown curriculum to replace the existing one with the view to position the country as a bastion of innovation in the new century the Secretary of Education has embarked on introducing the STEM Curriculum to the Schools of Excellence. It is hoped that students coming through the STEM Education will not only gain productive employment in the knowledge-based economy but become creators, innovators and inventors of knowledge and ultimately commercialize their creativity through development of their intellectual property rights.

In addition, there are four core elements espoused in this syllabus, insofar as student development is concerned: Students are expected to gain knowledge and skills and apply them to solve every day problems whilst appreciating Chemistry as a fundamental field of science.

This STEM Chemistry Syllabus for Grade 11 constitutes ten (10) Thematic Area (Thema) and is intended for students enrolling for STEM Education. In this booklet only three (3) Thema are presented. By studying these Thema students are expected to gain the foundational knowledge needed in Chemistry.

The success of this new Curriculum is highly dependent on the Teachers. They play a pivotal role by being innovative and keeping abreast of new information on scientific research and innovations. The challenge for teachers of Chemistry is to engage students to learn realistic context for increased and better understanding and appreciate the relevance of Chemistry to human lives, including good health and environment.

In order to ensure Teachers discharge their responsibilities adequately this syllabus is accompanied by a Teacher's Guide and Teacher's Resource Book which provides sufficient teaching and assessment materials to help Teachers.

I commend and approve this syllabus as the official curriculum for Chemistry to be used for the Grade 11 STEM Education in the Schools of Excellence throughout Papua New Guinea.

DR UKE KOMBRA, PHD
Secretary for Education

INTRODUCTION

Chemistry is the study of matter and its composition, how they interact to form new compounds through chemical reactions and the energy they generate as a result of these interactions. The concepts taught in Chemistry help us to understand these interactions at the molecular and atomic levels. Every substance, whether naturally occurring or artificially produced, consists of one or more elements of the periodic table.

Since Chemistry is an experimental science, laboratory work becomes an essential part of the syllabus. Practical work can be used to enable students to investigate the properties and reactions of substances and to provide opportunities for students to learn and test the principles and concepts of Chemistry.

The Grade 11 Chemistry Syllabus entails ten (10) Thematic Areas (Thema) and these are listed below. Each Thema has a single Module and three topics. Over time different Modules will be introduced.

Since Chemistry is a highly specialized subject students undertaking Chemistry are expected to possess high levels of cognitive and numeracy competency, and a firm grasp of the language skills.

The modules in this Syllabus are sequenced to guide teachers of Chemistry through the process of imparting basic chemistry knowledge and skills to students.

THEMA	MODULE	TOPICS
BIOTECHNOLOGY	12.1 Protein Sequencing	1. Amino Acids 2. Protein Structure and Function 3. Protein Purification 4. Protein Sequencing
HEALTH AND MEDICINE	12.1 Drug Development Process	1. Drug Targets 2. Drug Discovery 3. Drug Development and Production
ENERGY	12.1 Hydrogen – Future of Energy	1. Hydrogen as Fuel 2. Methods of Producing Hydrogen 3. Storing Hydrogen 4. Hydrogen Fuel Cell
BATTERY TECHNOLOGY	12.1 Solid-State Battery	1. Introduction to Solid-State Battery 2. Solid Electrolyte 3. Manufacturing process Solid-State Battery
ARTIFICIAL INTELLIGENCE	12.1 Artificial Molecular Machines	1. Introduction to Artificial molecular Machines 2. Catenanes and Rotaxanes 3. Other Protein based Motors under development
ROBOTICS	12.1 Robotics Meets Chemistry	1. Acidity Theories 2. Robotics 3. Programming a Robot

RATIONALE

There is a confluence of events that necessitated the need to introduce STEM Curriculum to the Schools of Excellence:

1. The School of Excellence Policy that was launched by Prime Minister Honorable James Marape, MP, in June 2020, calls for the establishment of the Schools of Excellence and the introduction of STEM Curriculum;
2. The Government has decided to replace the Observation Based Education to Standards Based Education in 2019 and tasked the Department of Education to introduce the STEM Curriculum as part of its reform agenda;
3. There is public outcry, both from public and private sectors, over the quality of graduates coming through the National Education System; and
4. The changing landscape in the technology domain globally and the pressing need to produce the next generation of workforce that is agile and can adopt to the changing circumstances as well as become enterprising individuals through their creation of intellectual properties.

The combination of these factors has led the Department of Education to adopt the STEM Curriculum and to introduce this to the Schools of Excellence as a necessary disruptive intervention in an effort to develop the next generation of national intellectual capital and to drive the growth and prosperity of the country through creation of intellectual properties.

In developing the STEM Curriculum, it was realized that the Grades 9 and 10 Science Syllabi does not provide a firm foundation to launch students into STEM Curriculum at the Schools of Excellence. Consequently, a STEM Bridging Program was necessary. The STEM Bridging Syllabus is a bridge across the divide that Grade 11 Students enrolling for STEM Education must cross.

In the Chemistry Bridging Syllabus students were introduced to the basic concepts and knowledge of Chemistry in order for them to develop an appreciation of the discipline in a broader context. In this syllabus students are introduced to advanced concepts in Chemistry as they are applied to the ten Thema.

The introduction of the STEM Curriculum, especially the Bridging Program and the Grades 11 and 12 Syllabi, will cause consequential changes to the Upper Classes of STEM Education and the Lower Secondary Syllabi. These syllabi will be introduced in the course of time.

STEM DOMAIN

At this juncture of our national development there is no other task before us more important than deploying a robust STEM Curriculum that develops the next generation of highly educated students to enter the work force. By building on the best of current practices and standards the STEM Curriculum aims to take us, beyond the constraints of the existing structures of schooling, towards a shared vision of excellence nationally and marketing the nation globally in terms of Science, Technology, Engineering and Mathematics. In this STEM Curriculum, there are ten (10) Thema (Thematic Areas) introduced.

Science

It is the study of the natural world, including the laws of the nature associated with physics, chemistry and biology and the treatment or application of facts, principles, concepts or conventions associated with these disciplines. Science is both a body of knowledge that has been accumulated over time and a process-scientific inquiry-that generates new knowledge. Knowing from science informs the engineering design process. In this STEM Curriculum the Science Domain has three Phyla – Biology, Chemistry and Physics.

Technology

Comprises the entire system of people and organizations, knowledge, processes, and devices that go into creating and operating technological artifacts, as well as the artifacts themselves. Throughout history humans have created technology to satisfy their wants and needs. Much of modern technology is the product of science and engineering and technological tools used in both fields. There are two Phyla – Information Technology and Computer Sciences – introduced in this STEM Curriculum for the Technology Domain.

Engineering

Concerns both the knowledge about the design and creation of products and a process for solving problems. The design process must result in a final product that must operate or function under various constraints. One constraint in engineering design is the laws of nature. Engineers must learn to harness the natural forces at work around them and through their design process develop products that must operate in harmony with nature. Other constraints include things such as time, money, available materials, ergonomics, environmental regulations and manufacturability. Engineering utilizes concepts in science and mathematics as well as technological tools. There are three (3) Phyla introduced in the Engineering Domain – Civil, Mechanical and Electrical.

Mathematics

It is the study of patterns and relationships among quantities, numbers, and shapes. Specific branches of mathematics include arithmetic, geometry, algebra, trigonometry and calculus. Mathematics is used in science and in engineering. In this STEM Curriculum, Mathematics is considered as a Service Domain; specific topics tailored towards specific Thema are also considered relevant to undergird the relevance and applicability of Mathematics Domain to other fields of knowledge.

STEM CHEMISTRY CONTENT OVERVIEW – GRADE 12, TERM 2

Table 1.1 Sequencing & Benchmarking STEM Chemistry Content Standards

Setting STEM Chemistry Content Standards for Progressive Learning & Assessment			
Thema	Energy	Module Content Standard	
Module 1 Hydrogen – Future of Energy		12.1a: Demonstrate an understanding of the energy potential of hydrogen as a fuel 12.1b: Explain how a hydrogen fuel cell works 12.1c: Produce Hydrogen through one of the Hydrogen productions processes 12.1d: Apply the concept of Hydrogen fuel to power vehicles	
Topics	Performance Indicators	Module Benchmark	Assessment Tasks
1. Hydrogen as Fuel	12.1.1a: Explain how hydrogen is an energy carrier 12.1.1b: Briefly explain the processes of producing hydrogen 12.1.1c: Discuss the pros and cons of Hydrogen as an energy carrier 12.1.1d: Discuss the challenge and future of Hydrogen as the future of energy	Student's master the knowledge and skills of: • Hydrogen as an energy carrier • Constructing a Hydrogen fuel cell • Producing Hydrogen via Electrolysis or solar process • Hydrogen fuels powering vehicles	Practical Assignment + Test + Quiz
2. Methods of Producing Hydrogen	12.1.2a: Explain how hydrogen is produce through the different methods 12.1.2b: Produce hydrogen using either the Direct Solar Water Splitting Processes or Electrolytic Processes		
3. Storing Hydrogen	12.1.3a: Explain how hydrogen is produced via Electrolysis (Water Splitting) 12.1.3b: Explain what an Electrolyzer is and the types of electrolyzers 12.1.3c: Explain and		

	produce hydrogen via the Solar Thermochemical hydrogen production process.		
4. Hydrogen Fuel Cell	12.1.4a: Construct a hydrogen fuel cell		

Thema	Battery Technology	Module Content Standard	
Module 1 Solid-State Battery		12.1a: Identify the difference between Lithium-Ion battery and Solid-State Battery 12.1b: Explain the future of battery technology with Solid-State Battery 12.1c: Explain the potential of Solid Electrolytes 12.1d: Explain the manufacturing process to construct a Solid-State Battery	
Topics	Performance Indicators	Module Benchmark	Assessment Tasks
1. Introduction to Solid-State Battery	12.1.1a: Differentiate between Lithium-ion vs. Solid-State Battery 12.1.1b: Explain why Lithium ion will be succeeded by Solid-State Battery 12.1.1c: Discuss the challenges that come with developing a Solid-State Battery	Student's master the knowledge and skills of: • Solid-State Batteries • Solid Oxide Electrolytes • Constructing a Solid-State Battery • Using Solid-State Batteries in Vehicles	Practical Assignment + Test + Quiz
2. Solid Electrolyte	12.1.2a: Explain what solid electrolytes are 12.1.2b: Explain how solid electrolytes battery work		
3. Manufacturing of Solid Batteries	12.1.3a: Describe the manufacturing process (Alternative A) 12.1.3b: Describe the manufacturing process (Alternative B) 12.1.3c: Describe the Cell Assembly and		

	Finishing Process 12.1.3d: Construct a simple battery		
--	--	--	--

Table 1.2 Mapping Content Progressions by Modules

Thema/Module	Module Content Standard	12.1	12.1
ENERGY / 12.1 Hydrogen – Future of Energy	12.1a: Demonstrate an understanding of the energy potential of hydrogen as a fuel 12.1b: Explain how a hydrogen fuel cell works 12.1c: Produce Hydrogen through one of the Hydrogen productions processes 12.1d: Apply the concept of Hydrogen fuel to power vehicles	✓	
BATTERY TECHNOLOGY / 12.1 Solid-State Battery	12.1a: Identify the difference between Lithium-Ion battery and Solid-State Battery 12.1b: Explain the future of battery technology with Solid-State Battery 12.1c: Explain the potential of Solid Electrolytes 12.1d: Explain the manufacturing process to construct a Solid-State Battery		✓

THEMA: ENERGY

Introduction

Energy is the ability to do work. It comes in several forms. It may exist in potential, kinetic, thermal, electrical, chemical, nuclear, or other forms of energy. However, the focus will be mainly on chemical energy in chemical reactions. In chemical bonds, the energy is stored in the bonds of chemical compounds. During a chemical reaction, chemical energy may be taken in or given off often in the form of heat.

Reactions that give off energy during a chemical reaction are said to be exothermic reactions while the ones that take in energy from the surrounding in order for the reaction to proceed are endothermic reactions. A good advantage of energy is that it can be transferred from a state to another without being destroyed. Some conversion of energy include the transfer of energy in food that we take in which is converted to mechanical energy and heat by the body, coal to electrical energy to supply electricity and conversion of chemical energy in batteries to supply energy through the process of electrolysis.

We can all agree that the sooner we move away from fossil fuels and toward other energy sources, the better. Two of the main reasons to move away includes Environmental pollution caused by non-renewable energy as well as the inevitability of the exhaustion of non-renewable energy sources. Therefore, we must look for ways to utilize the potential of renewable energy.

Listed are few of the many career pathways in the field of the energy industry:

- Civil Engineer
- Solar Photovoltaic Installer
- Petroleum Engineer
- Environmental Scientist
- Financial Analyst
- Electrical Power Line Installer
- Electrical Engineer
- Geoscientist
- Oil and Gas Service Unit operators
- Wind Turbine Technician
- Power Plant Operator; etc.,

MODULE 12.1: HYDROGEN – FUTURE OF ENERGY

5 Weeks

Hydrogen will replace electricity as a major energy carrier in the future because it can be produced safely from renewable energy sources and is virtually pollution-free. It will also be used to power zero-emission vehicles, heat homes and offices, generate electricity, and fuel planes. This module introduces students to the concept of hydrogen as the future's energy source. Students must be able to design technology that will include hydrogen as a source of energy in the form of electricity or fuel.

The outline of this module is structured in the following manner:

Topic 1: Hydrogen as Fuel

Topic 2: Hydrogen Fuel Cells

Topic 3: Hydrogen Production Processes

Topic 4: Application of Hydrogen Fuels - Vehicles

Linking Modules

Prerequisites:

- STEM-proper / Chemistry / Thema Energy / Module 11.1 Energy in Chemical Systems

Broad Content Standards

12.1a: Demonstrate an understanding of the energy potential of hydrogen as a fuel

12.1b: Explain how a hydrogen fuel cell works

12.1c: Produce Hydrogen through one of the Hydrogen production processes

12.1d: Apply the concept of Hydrogen fuel to power vehicles

Content Benchmarking

Students must master the knowledge and skills of:

- Hydrogen as an energy carrier
- Constructing a Hydrogen fuel cell
- Producing Hydrogen via Electrolysis or solar process
- Hydrogen fuels powering vehicles

Topic 1: Hydrogen as Fuel

Performance Indicators

Students are expected to:

- 12.1.1a:** Explain how hydrogen is an energy carrier
- 12.1.1b:** Briefly explain the processes of producing hydrogen
- 12.1.1c:** Discuss the pros and cons of Hydrogen as an energy carrier
- 12.1.1d:** Discuss the challenge and future of Hydrogen as the future of energy

Content Expansion

Body of Knowledge for Understanding

- Hydrogen as Fuel
 - Hydrogen as Energy Carrier
 - H₂ Carriers
- Introduction to Hydrogen economy
 - Rationale
 - Hydrogen Based Economy
 - Elements of a Hydrogen Based Economy
 - Comparison between Hydrogen Energy and Others
 - Technological Hurdles of Hydrogen Economy

Set of Proficiency Skills

- Analytical problem solving
- Research Skills
- Numerical Skills
- Written and Oral Communication skills
- Observational Skills
- Planning Skills
- Teamwork and interpersonal skills

Traits of Attitudes and Values

- Patience
- Determination
- Flexibility
- A logical and independent mind
- Meticulous attention to detail and accuracy

Topic 2: Methods of Producing Hydrogen

Performance Indicators

Students are expected to:

12.1.2a: Explain how hydrogen is produced through the different methods

12.1.2b: Produce hydrogen using either the Direct Solar Water Splitting Processes or Electrolytic Processes

Content Expansion

Body of Knowledge for Understanding

- Color-coding Hydrogen Production Methods
- Thermochemical Processes
 - Natural Gas Reforming
 - Steam Methane Reforming (SMR)
 - Partial Oxidation
 - Biomass Gasification
 - What is Biomass
 - How Does Biomass Gasification Work?
 - Biomass-derived Liquid Reforming
 - How does it work
 - Thermochemical water Splitting
 - Process
- Electrolytic Processes
 - How does it work?
 - Types of electrolyzers
 - Polymer Electrolyte Membrane Electrolyzers
 - Alkaline Electrolyzers
 - Solid Oxide Electrolyzers
- Direct Solar-Water Splitting Processes
 - Photoelectrochemical (PEC)
 - How does it work?
 - Why should we consider this method?
- Where does hydrogen come from?
- Where will the electricity for the production process come from?
- Prospects for the Future

Set of Proficiency Skills

- Analytical problem solving
- Research Skills
- Numerical Skills
- Written and Oral Communication skills
- Observational Skills
- Planning Skills
- Teamwork and interpersonal skills

Traits of Attitudes and Values

- Patience
- Determination
- Flexibility
- A logical and independent mind
- Meticulous attention to detail and accuracy

Topic 3: Storing Hydrogen

Performance Indicators

Students are expected to:

- 12.1.3a:** Explain how hydrogen is produced via Electrolysis (Water Splitting)
- 12.1.3b:** Explain what an Electrolyzer is and the types of electrolyzers
- 12.1.3c:** Explain and produce hydrogen via the Solar Thermochemical hydrogen production process.

Content Expansion

Body of Knowledge for Understanding

- Storing and Transporting Hydrogen
- Three different ways to store hydrogen:
 - Under High Pressure
 - In liquid form under cryogenic temperatures
 - On surface of or within solid and liquid materials.
- Materials-Based Hydrogen Storage
 - Underground Hydrogen Storage
 - Gas Grid Hydrogen Storage
- Why is Hydrogen Storage Important?
- Why is it difficult to store hydrogen?
- Is Hydrogen Storage Safe

Set of Proficiency Skills

- Analytical problem solving
- Research Skills
- Numerical Skills
- Written and Oral Communication skills
- Observational Skills
- Planning Skills
- Teamwork and interpersonal skills

Traits of Attitudes and Values

- Patience
- Determination
- Flexibility
- A logical and independent mind
- Meticulous attention to detail and accuracy

Topic 4: Hydrogen Fuel cell

Performance Indicators

Students are expected to:

12.1.4a: Construct a hydrogen fuel cell

Content Expansion

Body of Knowledge for Understanding

- Hydrogen Fuel Cell
 - What is a fuel cell?
 - Working Principle of a Fuel Cell
- Types of Fuel Cells
 - Polymer Electrolyte Membrane (PEM) Fuel Cell
 - Phosphoric Acid Fuel Cell
 - Solid Acid Fuel Cell
 - Alkaline Fuel Cell
 - Solid Oxide Fuel Cell
- Components of hydrogen fuel cells
- Advantages
- Applications
- Challenges of Hydrogen Electric Car (EC)
 - How an EC uses Hydrogen Fuel Cells
 - Energy Consumption
 - Example

Set of Proficiency Skills

- Analytical problem solving
- Research Skills
- Numerical Skills
- Written and Oral Communication skills
- Observational Skills
- Planning Skills
- Teamwork and interpersonal skills

Traits of Attitudes and Values

- Patience
- Determination
- Flexibility
- A logical and independent mind
- Meticulous attention to detail and accuracy

THEMA: BATTERY TECHNOLOGY

Introduction

A battery is a storage device and converter for chemical energy and electrical energy. It has one or more electrochemical cells that are connected to external circuits. The resulting conversion of the stored chemical energy to electrical energy powers the electrical devices such as flashlights, electric cars and mobile phones. When looking at the battery, the positive terminal is known as the cathode while the negative terminal is known anode.

Electrons flow from the anode then to the external circuit and onto the cathode. When the battery is connected in a circuit to give electricity, there is a reaction where electrons are gained and lost and this reaction is known as the reduction-oxidation reaction or in short, the redox reaction. The redox reaction in the battery converts the stored high chemical energy which reacts to low energy products and the free energy difference between the high energy of the reactants and the low energy of the products results as electrical energy to the external circuit.

Listed below are few of the many career pathways for students who are interested in the field of battery technology development.

- Cell Developer
- Process Engineering Technician
- Mechanical Engineer (Battery Performance Testing)
- Battery Product Design Engineer – Cell Mechanical Characterization
- Electrode Coating Process Chemistry/Engineer
- Battery Cell Engineer, and etc.

MODULE 12.1: SOLID-STATE BATTERY

5 Weeks

Battery technology could be a game-changer in the energy transition, allowing for the decarbonization of the transportation sector while also providing a necessary backstop for power generation from intermittent solar and wind sources. However, the widely used lithium-ion battery may not be up to the task of powering the global green economy's future.¹ Solid-state batteries have been touted as a long-term replacement for Li-ion batteries. Although research is continuing and prototypes are being developed, it could be a decade before a solid-state device is commercially available.

This module aims to encourage students to take interest in the development of solid-state battery as it is the future of battery technology. The outline of this module is structured in the following manner:

Topic 1: Introduction to Solid-State Battery

Topic 2: Solid Oxide Electrolyte

Topic 3: Building a Solid-State Battery

Topic 4: Application of Solid-State Battery - Vehicles

Linking Modules

Prerequisites:

- STEM-proper / Thema Battery Technology / Module 11.1 Chemistry Concepts of Batteries
- STEM-proper / Thema Battery Technology / Module 11.1 Physics of Batteries

Broad Content Standards

12.1a: Identify the difference between Lithium-Ion battery and Solid-State Battery

12.1b: Explain the future of battery technology with Solid-State Battery

12.1c: Explain the potential of Solid Electrolytes

12.1d: Explain the manufacturing process to construct a Solid-State Battery

Content Benchmarking

Student's master the knowledge and skills of:

- Solid-State Batteries
- Solid Oxide Electrolytes
- Constructing a Solid-State Battery
- Using Solid-State Batteries in Vehicles

¹ <https://www.forbes.com/sites/arielcohen/2021/02/11/what-batteries-will-power-the-future/?sh=5ce1b64841c0>

Topic 1: Introduction to Solid-State Battery

Performance Indicators

Students are expected to:

- 12.1.1a:** Differentiate between Lithium-ion vs. Solid-State Battery
- 12.1.1b:** Explain why Lithium ion will be succeeded by Solid-State Battery
- 12.1.1c:** Discuss the challenges that come with developing a Solid-State Battery

Content Expansion

Body of Knowledge for Understanding

- Lithium -ion battery vs. Solid-State Battery
 - Components
 - Li-ion Battery
 - Solid State Battery
- Reasons to develop Solid-State Battery
- How a Solid-State Battery Works
- Challenges

Set of Proficiency Skills

- Analytical problem solving
- Research Skills
- Numerical Skills
- Written and Oral Communication skills
- Observational Skills
- Planning Skills
- Teamwork and interpersonal skills

Traits of Attitudes and Values

- Patience
- Determination
- Flexibility
- A logical and independent mind
- Meticulous attention to detail and accuracy

Topic 2: Solid Electrolyte

Performance Indicators

Students are expected to:

- 12.1.2a:** Explain what solid electrolytes are
- 12.1.2b:** Explain how solid electrolytes battery work

Content Expansion

Body of Knowledge for Understanding

- Electrolyte
 - Oxide-based electrolytes
 - Sulfide-based electrolytes
- Materials
 - Anode
 - Cathode
- Chances and Challenges

Set of Proficiency Skills

- Analytical problem solving
- Research Skills
- Numerical Skills
- Written and Oral Communication skills
- Observational Skills
- Planning Skills
- Teamwork and interpersonal skills

Traits of Attitudes and Values

- Patience
- Determination
- Flexibility
- A logical and independent mind
- Meticulous attention to detail and accuracy

Topic 3: Manufacturing of Solid Batteries

Performance Indicators

Students are expected to:

- 12.1.3a:** Describe the manufacturing process (Alternative A)
- 12.1.3b:** Describe the manufacturing process (Alternative B)
- 12.1.3c:** Describe the Cell Assembly and Finishing Process
- 12.1.3d:** Construct a simple battery

Content Expansion

Body of Knowledge for Understanding

- Manufacturing Process – Electrode and Electrolytic Production – Alternative A
 - Compounding: Cathode and Electrode Production
 - Co-Extrusion: Cathode and Electrode Production
 - Extrusion and Calendaring: Anode Production
 - Laminating: Layer Compound Production
- Manufacturing Process – Electrode and Electrolytic Production – Alternative B
 - Grinding and Mixing
 - Layer Compound Production: High Frequency Sputting
 - Layer Compound Production: Sintering
 - Layer Compound Production: Thermal Evaporation
- Manufacturing Process – Cell Assembly and Cell Finishing
 - Cell Assembly: Cutting
 - Cell Assembly: Stacking
 - Cell Assembly: Contacting and Packaging
 - Cell Finishing: Formation
 - Cell Finishing: Aging
 - Cell Finishing: Testing and Grading

Set of Proficiency Skills

- Analytical problem solving
- Research Skills
- Numerical Skills
- Written and Oral Communication skills
- Observational Skills
- Planning Skills
- Teamwork and interpersonal skills

Traits of Attitudes and Values

- Patience
- Determination
- Flexibility
- A logical and independent mind
- Meticulous attention to detail and accuracy

REFERENCES

- [Masterdokument Zellherstellung v63 Vordruckfreigabe.pptx \(rwth-aachen.de\)](#)
- [Robot Programming: A Guide to Programming Your Own Robots \(edgy.app\)](#)
- [Nanorobotics \(fractal.org\)](#)
- [Making and Operating Molecular Machines: A Multidisciplinary Challenge - Baroncini - 2018 - ChemistryOpen - Wiley Online Library](#)
- [Driving Smart Molecular Systems by Artificial Molecular Machines \(wiley.com\)](#)
- [Masterdokument Zellherstellung v63 Vordruckfreigabe.pptx \(rwth-aachen.de\)](#)
- [<https://www.scienceinschool.org/article/2016/chembot-chemistry-robots/>](#)
- [Robot Programming: A Guide to Programming Your Own Robots \(edgy.app\)](#)
- [Robotics Designing of PCB - javatpoint](#)
- [<https://blog.finology.in/Legal-news/Ways-to-Commercialize-IPR>](#)
- [<https://www.upcounsel.com/why-do-patents-expire>](#)
- [<https://courses.lumenlearning.com/wmopen-introductiontobusiness/chapter/intellectual-property-2/>](#)
- [<https://www.forbes.com/sites/arielcohen/2021/02/11/what-batteries-will-power-the-future/?sh=5ce1b64841c0>](#)
- [<https://view.officeapps.live.com/op/view.aspx?src=https%3A%2F%2Fcourses.mak.ac.ug%2Fsites%2Fdefault%2Ffiles%2Fdownloads%2FIndustrial%2520Biochemistry.doc&wdOrigin=BROWSELINK>](#)
- [<https://www.biochemistry.org/education/careers/becoming-a-bioscientist/what-is-biochemistry/>](#)
- [<https://chemintelligence.com/ai-for-chemistry#:~:text=Artificial%20Intelligence%20%28AI%29%20is%20being%20used%20more%20and, and%20the%20time%20to%20market%20for%20new%20drugs.>](#)
- [<https://www.javatpoint.com/facebook-data-mining>](#)